

Proposed Solution

To address the challenges of the manual credit card approval process, a Machine Learning-based Credit Card Approval Prediction System is proposed. The system analyzes applicant information such as income, employment status, credit history, marital status, age, and other relevant attributes to predict whether a credit card application should be approved or rejected.

The proposed solution involves collecting and preprocessing the dataset, performing exploratory data analysis (EDA), and applying feature engineering techniques to improve data quality. Multiple machine learning algorithms, including Logistic Regression, Decision Tree, and Random Forest, are trained and evaluated to identify the model with the best prediction accuracy. The selected model is then integrated with a Flask-based web application, allowing users to enter applicant details through a simple HTML interface and receive instant prediction results. This automated approach reduces manual effort, improves consistency, and supports faster decision-making.

Expected Outcomes

The expected outcome of this project is to develop an efficient and reliable Credit Card Approval Prediction System that accurately predicts the approval status of credit card applications. The system is expected to reduce the time

required for processing applications, minimize human errors, and improve the consistency of approval decisions.

By comparing different machine learning models, the project aims to identify the most effective algorithm for prediction. The web-based interface provides an easy-to-use platform where users can enter applicant details and obtain immediate results. Overall, the project enhances the efficiency of the credit card approval process, supports better decision-making for financial institutions, and improves customer satisfaction by providing faster and more accurate predictions.